USER MANUAL – FACIAL EMOTION DETECTION SYSTEM

1. Introduction

The Facial Emotion Detection System is a Python-based application designed to help marketing teams analyse emotional reactions to advertisements. By processing facial images, the system automatically identifies the dominant emotion and provides confidence scores, categorized outputs, and logs for later analysis.

2. System Requirements

Hardware

- Windows, macOS, or Linux
- 4 GB RAM minimum
- 1 GB free storage

Software

- Python 3.8+
- Required libraries (installed via requirements.txt)

3. Installation

Step 1 — Download the Project

Clone the GitHub repository.

Step 2 — Open in PyCharm

Open the folder PythonProject1.

Step 3 — Install Dependencies

Code

```
pip install -r requirements.txt
```

4. Running the Program

Run:

Code

python main.py

You will see the main menu:

Main Menu

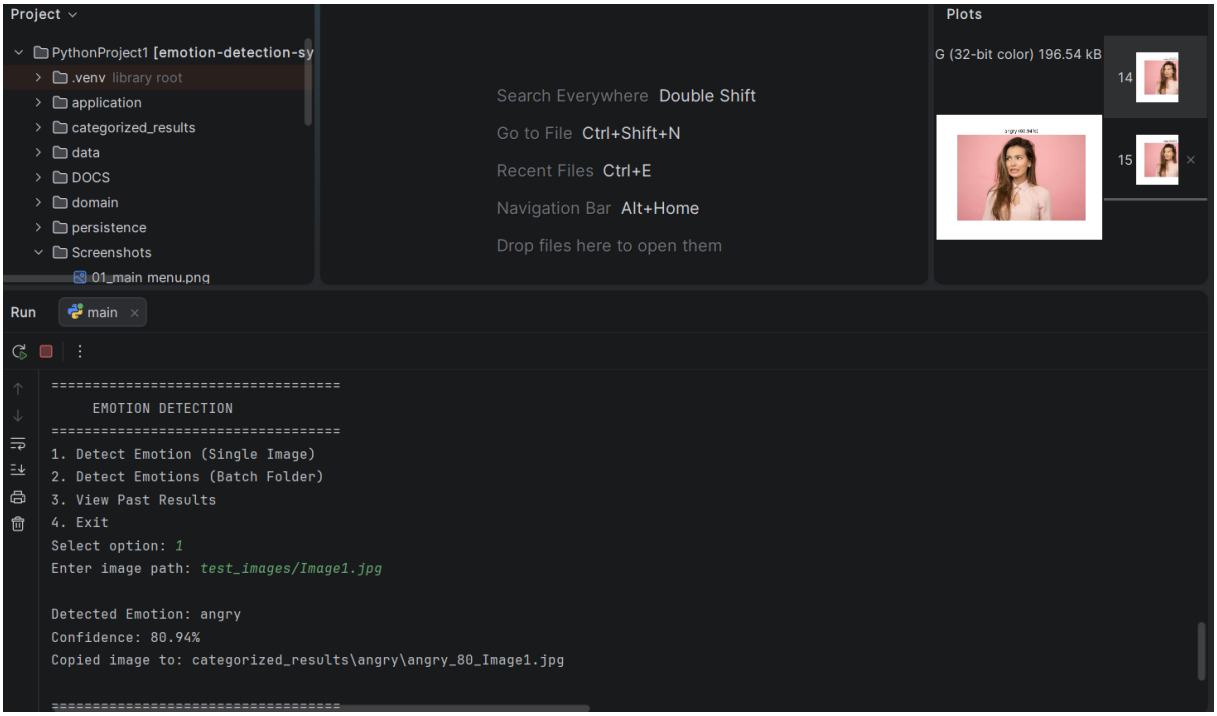
```
=====================================
     EMOTION DETECTION
=====================================
1. Detect Emotion (Single Image)
2. Detect Emotions (Batch Folder)
3. View Past Results
4. Exit
Select option:
```

5. Detect Emotion (Single Image)

How it works

- Loads one image
- Detects emotion
- Displays result
- Logs to CSV
- Copies image to categorized folder

Single Image Detection Output

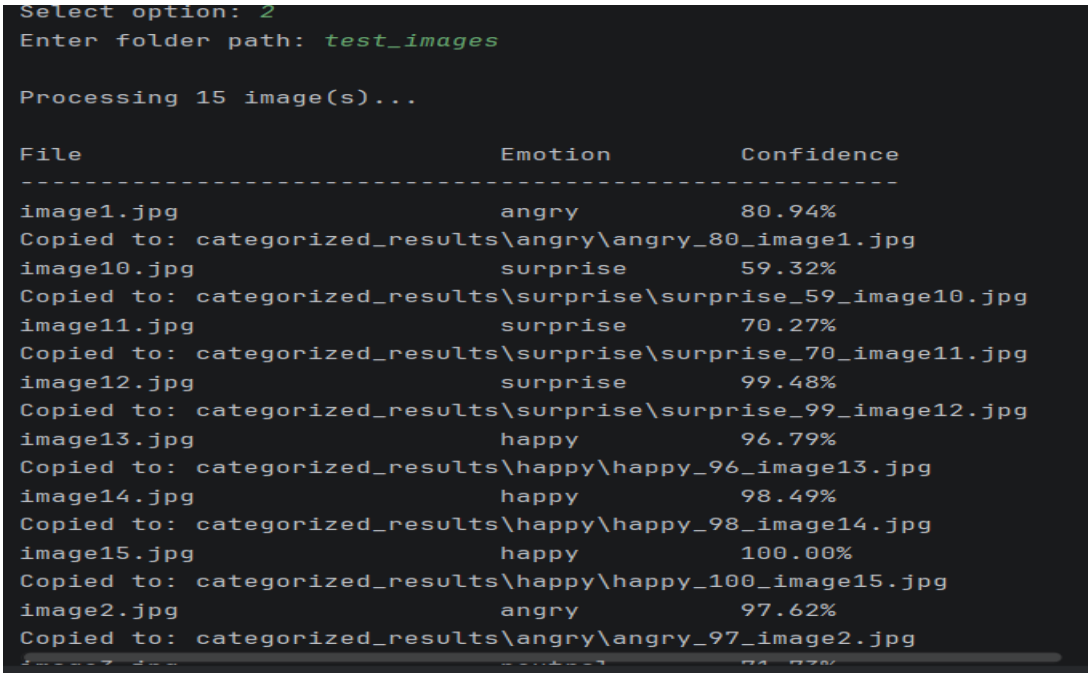


6. Detect Emotions (Batch Folder)

How it works

- Processes all images in a folder
- Prints results in a table
- Logs everything
- Categorizes images

Batch Processing Output



7. Viewing Past Results

Selecting option **3** displays all previous detections from the CSV log.

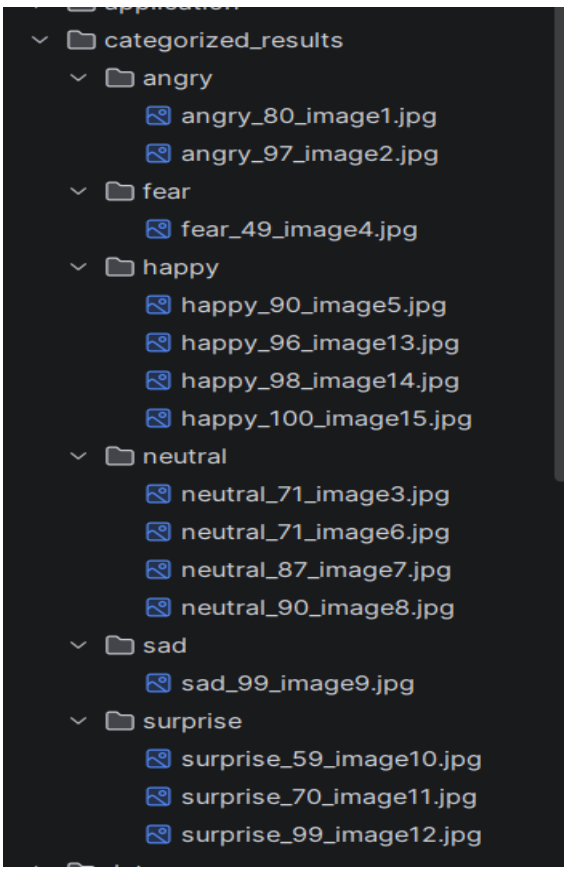
```
4. Exit
Select option: 3

Date & Time           Image                            Emotion        Confidence
-------------------------------------------------------------------------------------
2026-04-07 14:58:45   test_images/image3.jpg           neutral        71.73280
2026-04-07 15:13:34   test_images/image5.jpg           happy          90.51619
2026-04-07 15:13:48   test_images/image6.jpg           neutral        71.16679
2026-04-07 15:14:00   test_images/image7.jpg           neutral        87.20723
2026-04-07 15:14:11   test_images/image8.jpg           neutral        90.64640
2026-04-07 15:14:21   test_images/image9.jpg           sad            99.79858
2026-04-07 15:14:33   test_images/image10.jpg          surprise       59.31985
2026-04-07 15:14:42   test_images/image11.jpg          surprise       70.27487
2026-04-07 15:14:52   test_images/image12.jpg          surprise       99.47509
2026-04-07 15:15:01   test_images/image13.jpg          happy          96.78572
```

8. Output Files

1. categorized_results/

Contains emotion-based folders.

Categorized Results Folder Structure



2. results.csv

Stores all detections.

9. Supported Emotions

- Angry
- Happy
- Sad
- Neutral
- Surprise
- Fear
- Disgust

10. Troubleshooting

Problem: File not found

Check the path.

Problem: No face detected

Use a clearer image.

Problem: Performance issues

TensorFlow may show CPU optimization warnings.

Performance Test Output

```
✔ 1 test passed   1 test total, 3 sec 526 ms
C:\Users\mista\PycharmProjects\PythonProject1\.venv\Scripts\python.exe "C:/Program Files/JetBrains/PyCharm 2025.3.3/plugins/py
Testing started at 01:20 ...
Launching pytest with arguments C:\Users\mista\PycharmProjects\PythonProject1\tests\test_performance.py --no-header --no-summa

============================= test session starts ==============================
collecting ... collected 1 item

tests/test_performance.py::test_performance

============================== 1 passed in 15.61s ==============================
PASSED                      [100%]
Images processed: 15
Total time: 3.5220630168914795
Average time per image: 0.23480420112609862
2026-03-15 01:20:31.812774: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use av
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with t

Process finished with exit code 0
```

11. Maintenance

To update dependencies:

Code

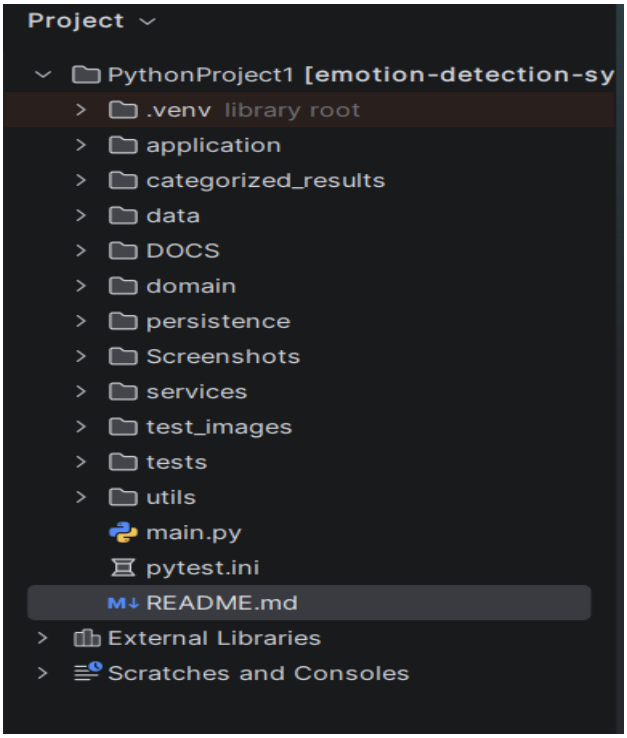
```
pip install --upgrade deepface tensorflow opencv-python
```

12. Future Features

- GUI
- Web interface
- Real-time webcam detection
- Database storage
- Multi-face detection

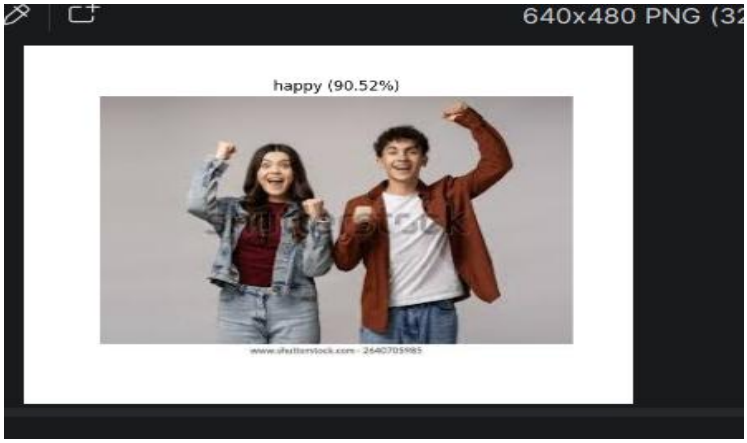
13. Appendix – Additional Screenshots

Project Structure



Single photo: different emotions

```
Select option: 1
Enter image path: test_images/image16.jpeg
2026-03-20 10:18:17.867372: I tensorflow/core/platform/cpu_f
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX

Detected Emotion: sad
Confidence: 76.49%
Copied image to: categorized_results\sad\sad_76_image16.jpeg
```

Single photo: same emotions

```
Select option: 1
Enter image path: test_images/image5.jpg
2026-03-20 10:12:21.665013: I tensorflow/core/platform/cpu_featu
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX

Detected Emotion: happy
Confidence: 90.52%
Copied image to: categorized_results\happy\happy_90_image5.jpg
```